

NAM DINH, Viet Nam

WMO: 488230

Lat: 20.4333N

Lon: 106.1570E

Elev: 3

StdP: 101.29

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.9	10.9	4.1	5.1	14.1	5.9	5.7	13.5	6.6	16.2	6.1	16.1	2.6	340	0.294

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	5.9	35.7	27.7	34.6	27.7	33.7	27.6	28.8	32.6	28.5	32.2	28.2	31.8	2.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.0	24.1	30.6	27.6	23.6	30.4	27.3	23.1	30.2	94.2	32.8	92.5	32.4	91.1	32.0	33.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.9	5.0	4.3	DB	8.4	37.6	1.5	1.1	7.2	38.3	6.3	39.0	5.5	39.6	4.3	40.3	(4)
			WB	6.9	30.0	1.5	0.8	5.8	30.6	4.9	31.1	4.1	31.5	3.0	32.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.5	17.1	18.4	20.6	24.8	28.1	30.1	30.0	29.1	28.0	26.0	22.6	18.7
	DBStd	5.34	3.62	3.87	3.12	2.82	2.37	1.87	1.79	1.66	1.84	2.24	2.95	3.22
	HDD10.0	1	1	1	0	0	0	0	0	0	0	0	0	0
	HDD18.3	166	68	45	14	1	0	0	0	0	0	0	3	34
	CDD10.0	5285	220	234	329	443	561	604	619	592	539	496	379	269
	CDD18.3	2409	30	46	85	194	302	354	361	333	289	238	132	45
	CDH23.3	24497	39	98	241	1355	3354	4654	4705	4044	3141	2002	732	131
	CDH26.7	9789	1	9	23	293	1339	2332	2289	1694	1122	548	129	10
Wind	WSAvg	2.0	2.0	1.9	1.8	2.1	2.2	2.1	2.2	1.8	1.8	1.9	1.9	1.8
Precipitation	PrecAvg	1651	29	23	51	67	174	180	246	304	317	161	69	29
	PrecMax	2775	176	51	127	203	376	427	414	549	706	507	236	93
	PrecMin	1167	1	4	5	20	64	90	83	133	120	28	6	2
	PrecStd	313	37	13	36	47	77	72	78	106	137	114	60	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.0	27.7	29.1	33.5	36.2	37.2	36.9	35.6	34.2	32.3	30.6	28.0
		MCWB	21.9	23.4	24.6	27.3	27.9	27.4	27.8	27.9	27.5	25.4	25.1	23.5
	2%	DB	24.4	25.6	27.0	30.9	34.4	35.9	35.3	34.2	33.0	31.4	29.4	26.2
		MCWB	21.0	22.4	23.7	26.4	27.9	27.6	27.8	27.9	27.2	25.1	24.4	22.3
	5%	DB	23.2	24.4	25.6	29.6	32.9	34.7	34.2	33.2	32.2	30.6	28.2	24.7
		MCWB	20.3	22.0	23.2	25.8	27.6	27.6	27.6	27.7	26.8	24.9	23.5	21.6
	10%	DB	22.0	23.4	24.6	28.3	31.7	33.5	33.2	32.1	31.2	29.6	27.0	23.4
		MCWB	19.9	21.5	22.7	25.3	27.1	27.6	27.4	27.5	26.4	24.6	22.9	20.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.9	23.9	25.1	27.8	29.0	29.4	29.2	29.2	28.7	27.1	26.1	24.5
		MCDB	24.7	26.2	27.7	32.0	33.6	33.3	33.4	32.5	31.9	30.1	28.6	26.6
	2%	WB	21.7	23.1	24.3	26.8	28.4	28.7	28.6	28.6	28.1	26.5	25.4	23.3
		MCDB	23.5	24.8	26.0	29.9	32.4	32.5	32.6	31.9	31.0	29.1	27.6	25.4
	5%	WB	21.0	22.5	23.7	26.2	28.0	28.4	28.2	28.2	27.6	26.0	24.9	22.4
		MCDB	22.7	24.0	25.2	28.8	31.7	32.1	32.0	31.5	30.4	28.5	27.0	24.1
	10%	WB	20.1	21.8	23.1	25.7	27.6	28.1	28.0	27.9	27.2	25.6	24.2	21.4
		MCDB	21.7	23.2	24.4	27.8	30.9	31.7	31.5	31.1	30.0	28.1	26.4	22.9
Mean Daily Temperature Range	5% DB	MDBR	4.7	4.2	4.1	4.8	5.8	5.9	5.4	5.2	5.2	5.7	5.8	5.6
		MCDBR	6.1	5.5	5.2	6.3	7.4	7.3	6.7	6.7	6.4	6.7	6.7	6.6
		MCWBR	3.0	2.6	2.7	2.9	2.8	2.2	2.2	2.3	2.0	1.9	2.3	3.0
	5% WB	MCDBR	5.2	4.8	4.6	5.9	6.7	6.4	6.1	6.0	5.8	5.8	5.8	5.9
		MCWBR	3.1	2.6	2.7	3.0	2.8	2.5	2.4	2.5	2.4	2.2	2.6	3.3
Clear-Sky Solar Irradiance	taub	0.613	0.646	0.809	0.763	0.562	0.513	0.484	0.539	0.599	0.707	0.613	0.643	
	taud	1.725	1.700	1.450	1.562	2.040	2.197	2.273	2.092	1.895	1.635	1.778	1.693	
	Ebn at Noon	662	674	588	627	761	792	816	776	724	620	657	618	
	Edh at Noon	217	235	313	282	173	147	136	164	197	247	204	219	
All-Sky Solar Radiation	RadAvg	2.35	2.73	3.00	3.98	5.05	5.22	5.02	4.61	4.36	3.88	3.36	2.60	
	RadStd	0.55	0.45	0.36	0.41	0.44	0.42	0.53	0.36	0.32	0.28	0.41	0.40	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.37	N/A	-0.93	+0.60	+0.21	+0.20	N/A	N/A	N/A	N/A
Regional (5 neighbors)		N/A	N/A	N/A	+0.55	+0.20	+0.21	N/A	N/A	N/A	+110

Nomenclature: See separate page